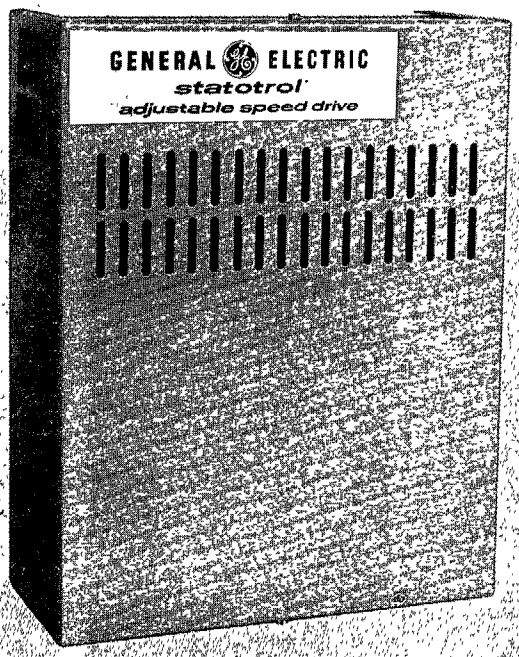
INSTRUCTIONS

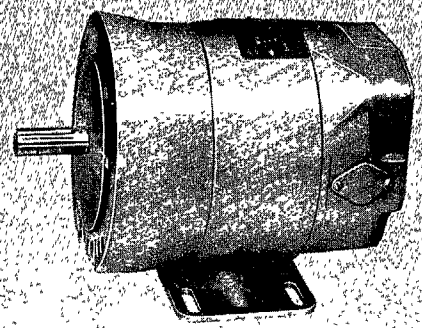
GEK-47612 A

STATOTROL* REGENERATIVE ADJUSTABLE SPEED DRIVE 6VHR . . . . . SERIES

1/6–2HP, SINGLE PHASE

115, 230 VOLTS, 50/60 HERTZ





DEPENDABLE MOTOR CONTROL

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company

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STATOTROL REGENERATIVE DC DRIVE

INTRODUCTION

This manual contains general information on the General Electric Statotrol regenerative DC drive. Additional information and diagrams may be supplied with the equipment when necessary.

The regenerative Statotrol is a high performance, electronic reversing controller intended for use in applications requiring rapid acceleration/deceleration, closed loop positioning or high start-stop duty cycles.

DESCRIPTION

The regenerative Statotrol drive consists of a controller and a DC motor. The controller contains forward and reverse SCR's, a loop contactor and the necessary electronic control circuitry. Built-in dynamic braking and thermal overload relay are available as modifications. A NEMA 1 cover encloses the controller.

The motor is either a shunt wound or permanent magnet field DC motor specially designed for use with a half wave phase controlled SCR power supply.

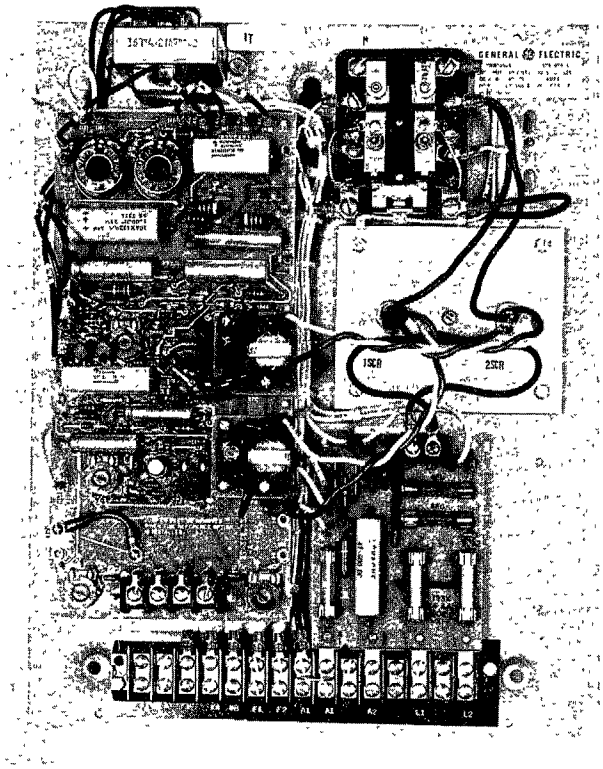


Figure 1. 6VHR Regenerative Reversing Statotrol

RECEIVING AND STORAGE

As soon as the equipment is received, it should be unpacked and examined for damage sustained in transit. If damage is evident, a damage claim should be filed immediately with the transportation company, and the local General Electric Co. Sales Office should be promptly notified.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected against accidental damage. Avoid storage in a location where construction work is in progress.

INSTALLATION

CONTROLLER INSTALLATION

The Statotrol 6VHR Control should be wall mounted in a location which will allow free flow of cooling air over the controller. Maximum ambient temperature around the controller must not exceed 40°C (104°F). Figures 2 and 3 give mounting dimensions and outline dimensions. It is recommended that two or more inches of clearance be provided all around the controller to assure adequate air flow for cooling. If the controller is mounted inside another enclosure, the controller cover may be discarded.

MOTOR INSTALLATION

The motor must be firmly mounted and properly aligned to prevent vibration. Excessive vibration causes rapid wear and audible noise. Heat dissipated by the motor will raise the ambient temperature around the motor if the motor is installed in an enclosed space. Since a lower ambient temperature will extend the motor life, it is always desirable to mount the motor in a well ventilated location. The motor ambient temperature should never exceed 40°C (104°F) unless an oversized motor has been specially selected.

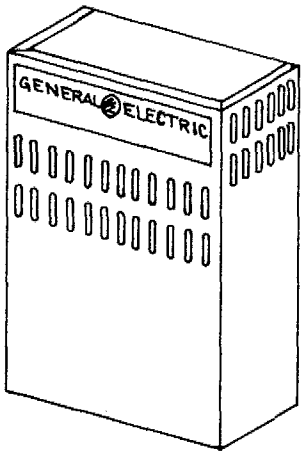
WARNING

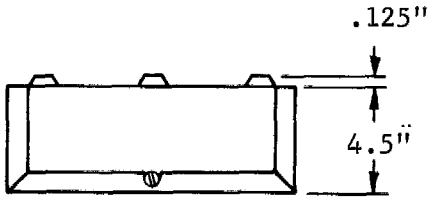
IF THE MOTOR IS ACCESSABLE WHILE IT IS RUNNING, A PROTECTIVE GUARD SHOULD BE INSTALLED AROUND ALL EXPOSED ROTATING PARTS.

ELECTRICAL CONNECTIONS

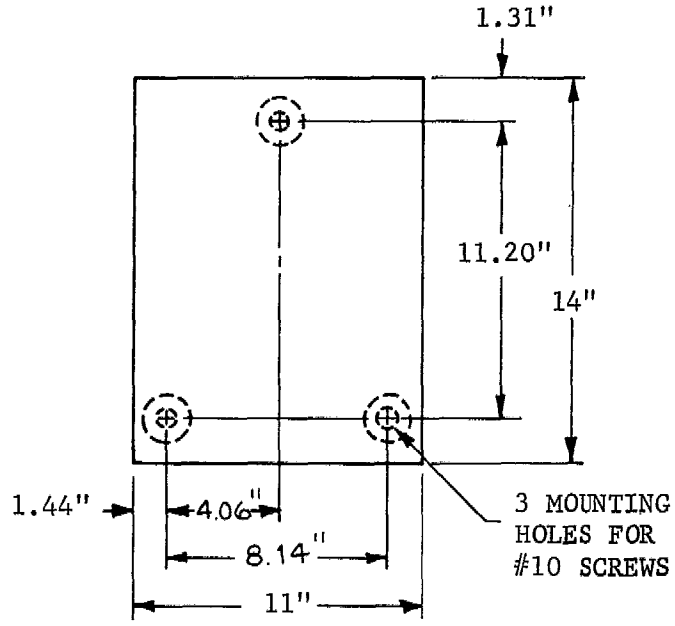
PROPER LINE VOLTAGE CONNECTION

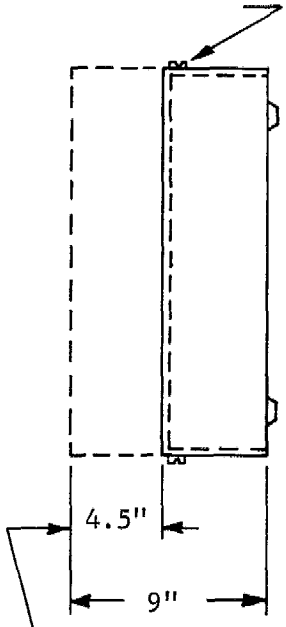
The controller operates on 50/60 cycle AC. Input volt-



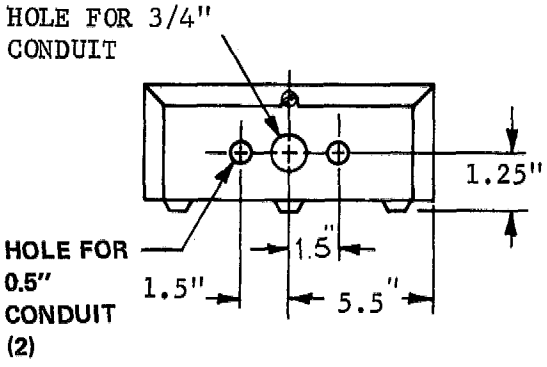


CLEARANCE REQUIRED TOP AND BOTTOM FOR ACCESS TO SCREWS





CLEARANCE REQUIRED TO REMOVE FRONT. (NOT REQUIRED WHERE COVER IS NOT USED).



age rating is either 115 volts or 230 volts, not both. Check the nameplate to determine proper input voltage. A transformer is required where supply voltage deviates more than 10% from the controller rating.

GROUNDING

WARNING

PERSONNEL SAFETY CONSIDERATIONS AND THE NATIONAL ELECTRIC CODE REQUIRE THAT ELECTRICAL APPARATUS ENCLOSURES BE SOLIDLY CONNECTED TO BUILDING GROUND.

The necessary wire gauge of the ground wire is determined by the rating of the branch circuit breaker. Connect a suitably grounded lead to the ground connection screw provided in the lower, right hand corner of the SCR mounting plate. Do not rely or depend on mounting bolts for a ground connection.

WARNING

PROPER MOTOR GROUNDING IS ESSENTIAL FOR PERSONNEL SAFETY.

Do not depend upon motor mounting bolts to ground the motor frame. The ground connection provided in the motor depends upon the type, size, and enclosure of the motor. Each Statotrol motor has one of the following provisions for connecting a ground lead.

1. A four inch green wire inside the wire well.

2. A 1/4-20 bolt in a tapped hole in the motor end shield.

3. A brass screw which serves as a ground terminal and also mounts the conduit box found on the side of TEFC motors.

4. A drilled hole in the motor end shield (inside the wire well near the electrical terminals) suitable for a self tapping screw.

If there is any doubt about motor lead identification, the problem can be resolved with a simple resistance check. The motor armature resistance will be about 0.3 to 15 ohms while field resistance will be about 75 to 500 ohms. There must be no continuity between armature and field. Permanent magnet DC motors have armature leads, but no field leads. 1-1/2 HP motors may have leads labeled C1 and

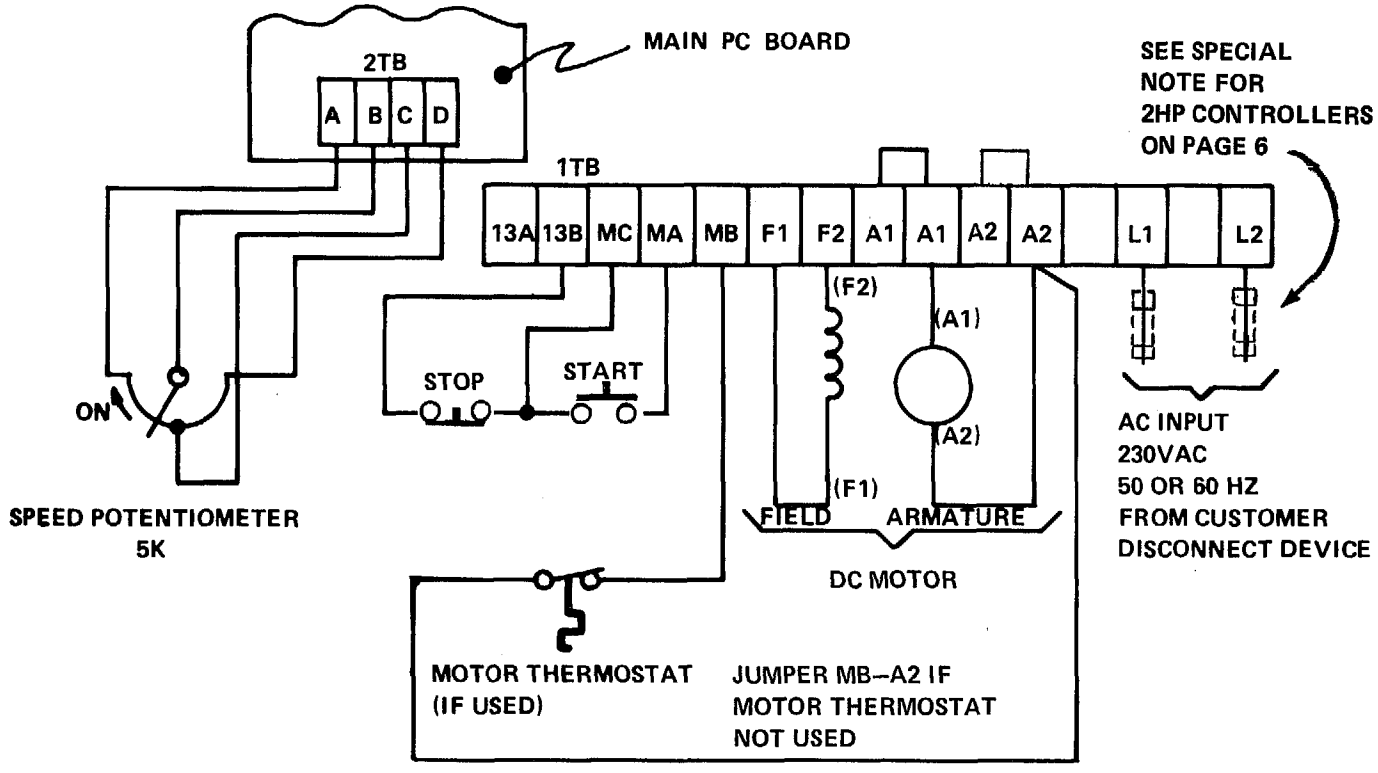


Figure 3. Customer Connection Diagram

C2. These leads should be individually taped, well insulated and left unconnected.

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. SINCE THE CONTROLLER STOP SWITCH DOES NOT REMOVE VOLTAGE FROM THE CONTROLLER OR THE MOTOR FIELD, A BRANCH CIRCUIT BREAKER OR A FUSED DISCONNECT SWITCH MUST BE USED TO DISCONNECT THE CONTROLLER WHENEVER SERVICE WORK IS TO BE PERFORMED ON THE DRIVE.

POWER WIRING

A 15 point terminal board, 1TB, is provided for connection to the AC power lines, motor leads, and control circuit. Reference (speed potentiometer) connections are made to the 4 point terminal board, 2TB on the main printed circuit board.

When the motor is connected as shown in Figure 3, a negative reference voltage (speed potentiometer CCW) will result in CCW rotation of the motor viewed from the shaft end. Reversing the motor armature leads or the reference polarity will reverse rotation.

SPECIAL NOTE FOR 2HP CONTROLLERS

2HP Controllers require external fusing for short circuit protection. The internal controller fuses are for SCR overload protection only, and are not suitable for interrupting short circuit currents. Recommended rating for the external fusing is 30 amp, 250V, 10,000 amp I.C.

CAUTION

A CONNECTION ERROR IN THE POWER WIRING CAN DAMAGE THE CONTROL. RECHECK THE CONNECTIONS BEFORE POWER IS APPLIED.

CAUTION

TO PREVENT ELECTRICAL NOISE PICKUP, TWISTED OR SHIELDED WIRE IS RECOMMENDED IN ALL REFERENCE AND FEEDBACK CIRCUITS.

SPECIAL CIRCUITS

HALF WAVE FIELD

Standard motors for the 6VHR Statotrol drive have full wave fields. This means that a motor with a 75 VDC armature has a 100VDC field and a motor with 140 VDC armature has a 200 VDC field.

Motors with half wave fields require half the field voltage of the standard motor i.e. 50 VDC and 100 VDC for 75 VDC and 140 VDC armatures respectively. These motors can be used with 6VHR controllers providing their fields are connected as in Figure 4 below.

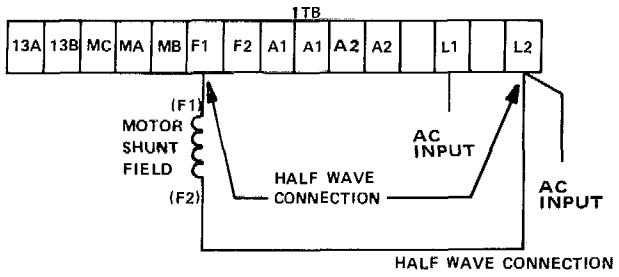


Figure 4. Half-Wave Field Connection

GROUNDED AC LINE

If the AC power line is grounded, the grounded lead must be connected to 1TB L2. In addition, 230 volt input controllers should have a jumper added 1TB L2 to 1TB A2. The jumper is desirable because it "shorts out" the fuse, 2FU, which would otherwise be between the motor armature (A2) lead and building ground. 2 FU can then be used as a spare fuse. See Figure 5.

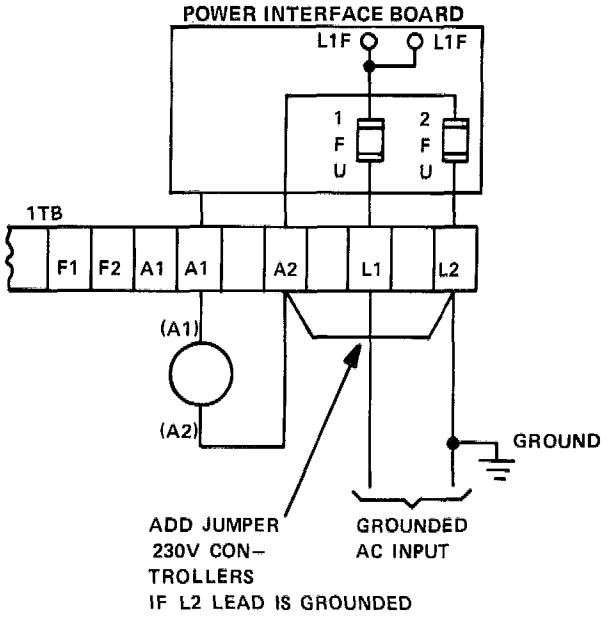


Figure 5. Grounded AC Line

TACHOMETER FEEDBACK

The 6VHR controller is compatible with most DC tachometer generators. All that is needed to obtain tachometer feedback is a DC tachometer generator, a No. 18 wire spade lug, and a tach voltage calibrating resistor (36R). See Figure 6. Note that the tachometer output voltage at rated speed must be equal to or greater than the motor armature voltage (either 75 or 140 VDC).

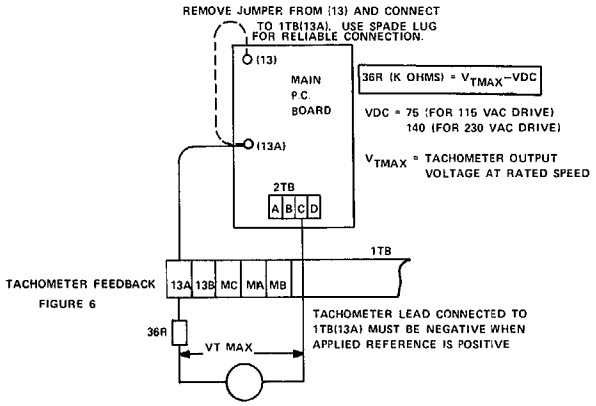


Figure 6. Tachometer Feedback

AUTO AND AUTO/MAN REFERENCE

Figures 8A and 8B give special connection schemes for using the reference signal from an automatic process controller.

An input signal of 8 volts and 1.6 ma is required to achieve rated motor speed.

If the voltage source output is greater than 8 volts, select the proper value of dropping resistor, 35R, using the formula given below. Connect the voltage source to the control as shown in Figure 8A or 8B.

$$35R\ (K\ OHMS) = \frac{VDC - 10V}{3.2\ MA}$$

The RMS ripple on the reference voltage must be less than 1%. If necessary, the customer must supply a capacitor across the voltage source to reduce the ripple to a value below 1%. The signal source should be isolated from ground. If an ungrounded signal source is not available, use an isolation transformer to supply AC power to the drive. A negative signal applied to terminal 2TB-B with respect to terminal 2TB-C will run the motor in a clockwise direction when viewed from the commutator end. A positive signal applied to terminal 2TB-B with respect to terminal 2TB-C will run the motor in a counterclockwise direction. Clockwise rotation is considered "forward" and counterclockwise rotation is considered "reverse".

INSTRUMENT FOLLOWER MODIFICATION

Current Source

The minimum required current input signal for rated motor speed is 1.6 milliamps. If the current source output is greater than 1.6 milliamps, select load resistor, 35R, using the formula given in the following equation.

$$35R\ (OHMS) = \frac{8000}{1MAX\ (ma) - 1.6}$$

Connect the current source in place of the voltage source shown in Figures 8A and 8B. Resistor 35R is then connected in parallel with the current source.

The RMS ripple must be less than 1%. If necessary, the customer should connect a capacitor across 35R to reduce the ripple to a value below 1%. The signal source should be isolated from ground. If an ungrounded signal source is not available, use an isolation transformer to supply AC power to the drive. A negative signal applied to terminal, 2TB-B, with respect to terminal 2TB-C will run the motor in a clockwise direction when viewed from the commutator end. A positive signal applied to terminal 2TB-B, with respect to terminal 2TB-C, will run the motor in a counterclockwise direction. Clockwise rotation is considered "forward" and counterclockwise rotation is considered "reverse."

INITIAL OPERATION AND ADJUSTMENT

CURRENT LIMIT ADJUSTMENTS

The Statotrol current limit circuitry works by limiting the turn-on time of the SCR's. This limits the effective voltage across the motor armature, and the armature impedance then limits the current. The Statotrol controller is factory-set to limit the current in a low impedance motor. If a high impedance motor is to be used, it may be necessary to adjust potentiometers 3P and 4P to get the desired motor torque output.

SET POTENTIOMETER 3P

Turning potentiometer 3P clockwise will increase the current which will flow in the armature when the rotor is locked. Since motor output torque is proportional to armature current, 3P effectively adjusts the starting torque of the motor.

How to Set 3P by Measuring Armature Current

Connect a DC ammeter in series with the motor armature, lock the rotor, turn the reference potentiometer in either direction, and energize the M relay. Turn potentiometer 3P until the desired armature current is achieved. Do not stall the motor for more than one (1) minute or the motor may be damaged.

CAUTION

DO NOT SET DC STALL CURRENT TO EXCEED 200% OF MOTOR NAMEPLATE-RATED CURRENT OR DAMAGE MAY RESULT. CONSULT THE FACTORY IF HIGHER TORQUE IS REQUIRED.

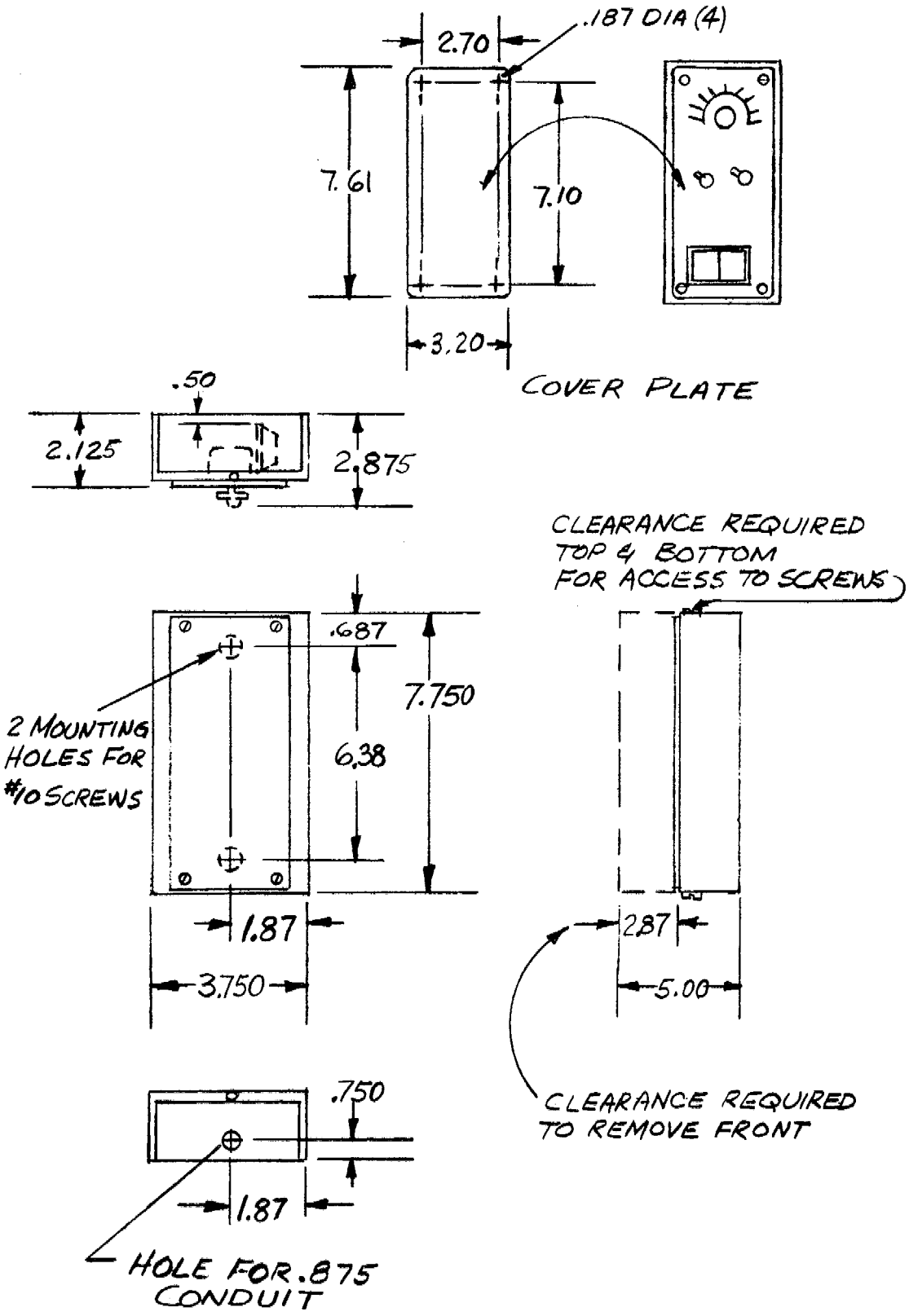


Figure 7. Remote Control Station Outline and Mounting Dimensions

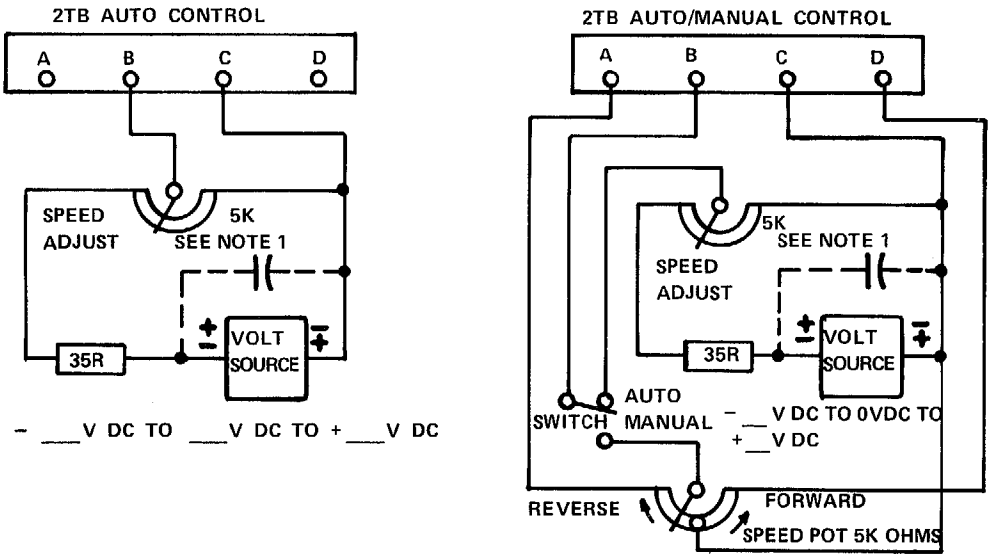
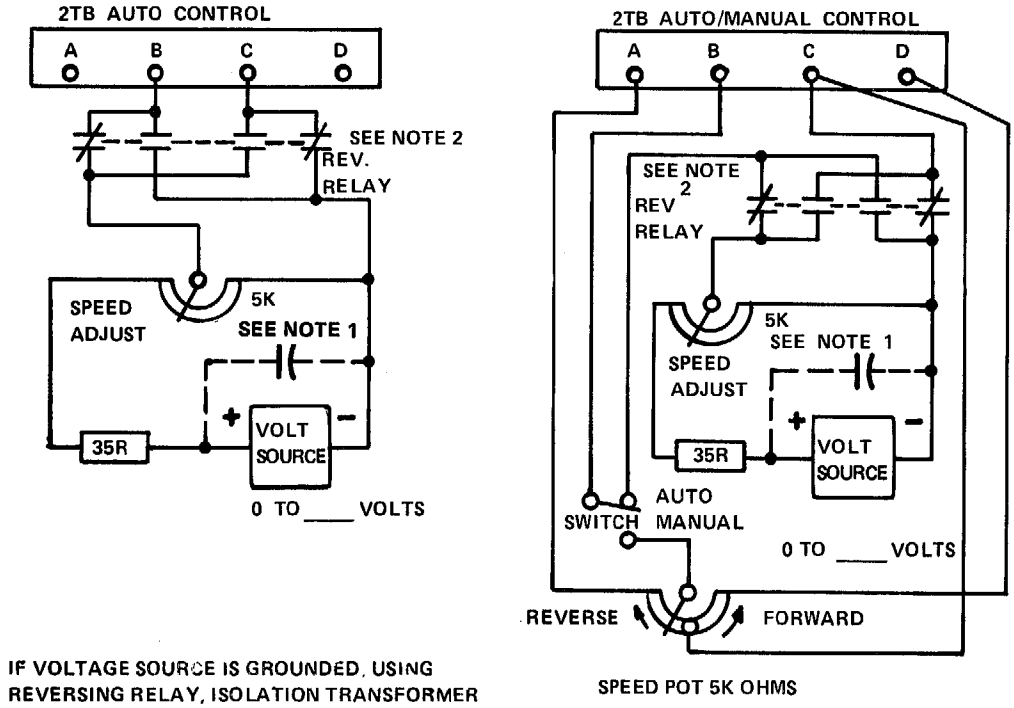


Figure 8A. Connections for a Reversing Polarity Voltage Source



IF VOLTAGE SOURCE IS GROUNDED, USING REVERSING RELAY, ISOLATION TRANSFORMER MUST BE USED.

Figure 8B. Connections for a Single Polarity Voltage Source

NOTE 1: CAPACITOR SUPPLIED BY CUSTOMER TO REDUCE VOLTAGE RIPPLE TO UNDER 1%

NOTE 2: RECOMMEND HI FIDELITY SEALED RELAY FOR RELIABILITY

How to Set 3P Without an Ammeter

An approximate setting for stall current can be achieved by aligning the correct ink spots on potentiometer 3P. Read the instructions carefully since turning 3P too far clockwise may result in damage to the motor or the controller. Consult the factory if the settings given below do not give adequate torque for the application.

To set 3P, find the proper motor in the left hand column of Table 1 and set 3P to the mark indicated in the right hand column.

SET POTENTIOMETER 4P

Turning 4P clockwise will increase the available motor output torque at rated motor speed.

How to Set 4P by Measuring Armature Current

With a DC ammeter in the armature circuit, lock the rotor, turn the reference potentiometer in either direction, and energize the M relay. Read the current in the armature, release the rotor, and observe the ammeter as the motor accelerates. Potentiometer 4P must be set so that the armature current holds steady or gradually decreases as the motor accelerates to top speed. Turning 4P clockwise will increase the armature current available at higher speeds. For smooth acceleration and long motor life, it is recommended that 4P be turned to the farthest counterclockwise setting which will allow desired motor performance.

How to Set 4P Without an Ammeter

An approximate setting for current limit can be achieved by aligning the correct ink spots on potentiometer 4P. Find the proper motor in the left hand column of Table I and set 4P to the mark indicated in the right hand column of Table I.

OVERLAP SETTING

Potentiometer 2P is the overlap adjustment. Turning 2P clockwise gives faster "breakaway" response and decreases the "deadband" as the reference signal goes from forward to reverse. Turning 2P clockwise also increases motor heating at low speed and may contribute to stability problems in certain applications. Therefore, 2P is set fully counterclockwise as a factory adjustment, and it is recommended that 2P be kept fully counterclockwise except in those applications which specifically require a narrow reference deadband between forward and reverse.

GAIN ADJUSTMENT

Turning 5P clockwise increases the gain of the drive system. High gain makes the system very responsive to transient changes in the motor load or the reference signal. As in any closed loop feedback system, high gain can contribute to system instability as well as transient response overshoot and ringing. Therefore, 5P is factory-set to minimum gain (full CCW) and it is recommended that 5P be kept in the minimum gain position except in those applications which specifically require high gain.

BALANCE ADJUSTMENT

Balance Potentiometer 7P is used to establish the reference voltage which will stop the motor. 7P is factory-adjusted by setting the reference signal to zero (jumpering 2TB-B to 2TB-C) and adjusting 7P until the motor stops. This is the proper setting for most applications.

In some applications it may be desirable to have the motor stop at some reference voltage other than zero. For these applications, set the reference signal to the voltage at which it is desired that the motor stop, and then turn 7P until the motor does stop.

TABLE I
CURRENT LIMIT SETTINGS

STATOTROL MOTORS THROUGH 2 HP	3P SETTING AND 4P SETTING
All 2500 RPM, TENV, TEFC	Counterclockwise dot
All 1725 RPM, TENV, TEFC and 2500 RPM, D. P.	Center dot
All 1725PRM, D. P.	Clockwise dot
TEFC = Totally enclosed, fan cooled TENV = Totally enclosed, non-ventilated D. P. = Drip-proof, open construction	

MAXIMUM SPEED ADJUSTMENTS

Turning 1P clockwise will increase the maximum reverse speed at which the motor will run. Turning 6P clockwise will increase the maximum forward speed at which the motor will run. Electrically, these potentiometers accomplish their function by limiting the range of the speed reference potentiometer in the control station.

Potentiometers 1P and 6P are factory-set to limit the motor speed to a value about equal to the motor nameplate rating. The customer can adjust these potentiometers by turning the reference signal potentiometer in the control station to one extreme and adjusting the maximum speed and then repeating the operation for the other extreme. It is recommended that motors never be operated at speeds above their nameplate rating.

TIMED REFERENCE OPTION ADJUSTMENTS

The timed reference circuit is available only as a factory-installed option. It consists of a small component board which is attached to the main component board by five screws. An auxiliary contact on the M relay is used to bypass the time delay and reset the reference to zero when the M relay is de-energized.

Turning the timed reference option potentiometer 101P clockwise will increase the time the motor take to accelerate from the one speed to another when the reference signal is changed from stop to any reverse speed, or is changed from any high forward speed to a lower forward speed or a reverse speed.

Similarly, turning 102P clockwise will increase the time the motor takes to accelerate from one speed to another when the reference signal is changed from stop to any forward speed or is changed from any high reverse speed to a lower reverse speed or any forward speed.

MAINTENANCE

MOTOR INSPECTION

Limited life items in the motor include the brushes, the commutator, the bearings, and the winding insulation. If the motor is not overloaded, the bearings and insulation should provide years of troublefree operation. However, the brushes and the commutator will wear with use. Factors which shorten brush life include high speed operation, high armature current, high armature voltage, extremely low humidity, and scratches or pits on the commutator surface. In severe duty applications the brushes should be inspected after every 1000 hours of operation. The brushes should be replaced anytime it is found that less than 1/2 inch of brush material remains. The bearings are packed with a special grease during assembly and should not be relubricated.

Make sure that the motor mounting bolts are all tight. A loose mounting bolt can cause considerable noise and may result in excessive wear.

If the motor requires maintenance, please refer to the General Electric Motor Maintenance Manual for the particular motor used on the application.

CONTROLLER INSPECTION

Inspect the contacts of the M relay, and replace the relay if the contacts are worn. Keep the heat sink clean, and do not allow anything to block the flow of air across the heat sink and through the enclosure.

TROUBLESHOOTING AND REPAIR

Many minor operational problems can be traced to improper adjustment of the various potentiometers on the component board. Proper adjustments are described in the "Potentiometer Adjustments" section of this book. Other possible problems are discussed below:

WARNING

SINCE THE CONTROLLER STOP BUTTON DOES NOT REMOVE VOLTAGE FROM THE CONTROLLER OR THE MOTOR FIELD, A BRANCH CIRCUIT BREAKER OR A FUSED DISCONNECT MUST BE USED TO DISCONNECT THE CONTROLLER WHENEVER WORK IS TO BE PERFORMED ON THE DRIVE.

If the motor will not run:

1. Make sure all connections are correct and tight. Check fuses 3FU, 4FU, and 5FU.

2. Make sure there is AC power available at the controller terminal. Check power switch closure.

3. Check the power fuses 1FU and 2FU. If a fuse is blown, replace it. If it blows again look for short circuit or motor overload.

CAUTION

1FU AND 2FU MUST BE HIGH INTERRUPTING CAPACITY, CERAMIC BODY, 250 VOLT FUSES, 1/4 INCH IN DIAMETER AND 1-1/4 INCHES LONG. RECOMMENDED FUSES ARE LITTLE-

FUSE 314000 SERIES, TYPE 3AB "NORMAL BLO" FUSES AND BUSSMANN TYPE ABC FUSES.

If new fuses also blow, refer to Table 1 and check the settings of 3P and 4P. With an analyzer set on the X1000 ohms scale, disconnect power and verify that the reading from the cathode to the anode of the SCR's is open circuit. If the reading is not open circuit, one or both of the SCR's may have failed short. Also check the motor armature and field leads for shorts to ground.

If the fuse is not blown, but the motor still will not run, check the field voltage from 1TB-F1 to 1TB-F2. The field voltage should be 100 volts DC and 200 volts DC for input power voltages of 115 volts or 230 volts respectively. If there is no field voltage, check the field power supply diodes which are mounted on the power interface board.

4. Check the motor for internal shorts. An ohmmeter should read 0.3 to 15 ohms from A1 to A2, and should read from 75 to several hundred ohms from F1 to F2.

Rotating the armature slowly while reading the armature resistance helps check for intermittent shorts in the motor.

If the motor runs faster or slower than normal for a given setting of the speed control:

1. Check the line voltage to be sure that it is normal, and check fuse 4FU.

2. Check the reference voltage to be sure that it is normal. If it is low, check the setting of the max speed potentiometers 1P and 6P.

3. Check the current limit adjustment. (See Table 1.)

4. Check for an overload on the motor. The motor is overloaded if the armature current exceeds the motor nameplate rating.

If the motor runs at full speed and cannot be controlled by speed control:

1. Check all connections for poor connection or short circuits. Check fuse 3FU.

2. Disconnect line power and check the SCR's with an ohmmeter set to the X1000 scale. The SCR's should read open circuit from the cathode to the anode and heat sink.

3. With the motor running, check the voltage from main printed circuit board terminal 13A to 2TB-C. This is the feedback signal and must be equal to armature voltage. If this voltage is not equal to armature voltage, disconnect the power and check continuity from point 13A to 1TB-A1.

If the motor will run in only one direction:

1. Verify that the reference voltage measured from 2TB-B to 2TB-C is correct. 2TB-B must be positive with respect to 2TB-C for reverse rotation and negative with respect to 2TB-C for forward rotation.

2. Firing circuit 2FC triggers 2 SCR to make the motor go forward. 1FC triggers 1SCR to make the motor go in the reverse direction. If the motor will not run in one direction, replace the corresponding firing circuit component board. If this does not solve the problem, replace the corresponding SCR.

If all previous troubleshooting steps have been unsuccessful:

1. Check the voltage on the transformer leads where the leads connect to the component board. The voltages should be as shown in Table 2. If any voltages are wrong, disconnect the control from the power line and disconnect the questionable transformer leads from the component board. Isolate the loose leads from all conductors, open circuit the M relay, connect the control to the power line and check the voltage again. If the voltage is now correct, the main component board should be replaced. If the voltage is still incorrect, replace transformer 1T.

2. Replace the main component board.

TABLE II
IT TRANSFORMER VOLTAGES

CIRCUIT NUMBERS	NOMINAL VOLTAGE
8 to 46	16 volts RMS ± 20%
8 to 48	16 volts RMS ± 20%
38 to 34	4 volts RMS ± 20%
38 to 53	4 volts RMS ± 20%

RENEWAL PARTS

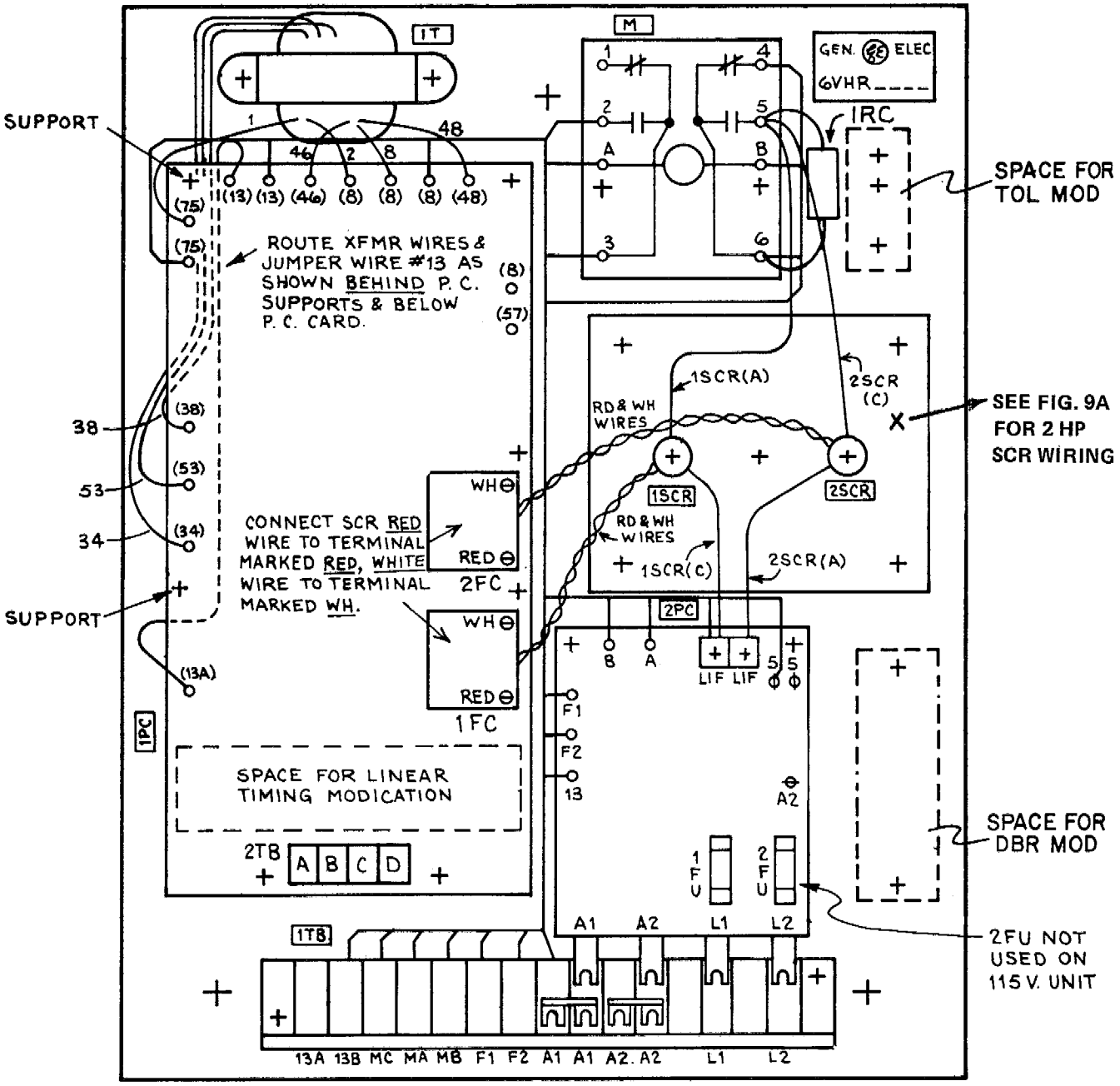
When ordering renewal parts, contact the nearest General Electric Sales Office, giving the quantity required and catalog numbers. In addition, give the 6VHR number and the complete nameplate rating of the control.

TABLE III

PRINCIPAL RENEWAL AND SPARE PARTS LIST

PART	CATALOG NUMBER	6VHR MODEL NUMBER												
		1016	1025	1033	1050	1075	2016	2025	2033	2050	2075	2100	2150	2200
IPC	193X638AAG05	1	1	1	1	1								
IPC	193X638AAG06						1	1	1	1	1	1	1	1
2PC	193X641AAG01	1	1	1	1	1								
2PC	193X641AAG02						1	1	1	1	1	1	1	1
1, 2 SCR	942B279CAG02													2
1, 2 SCR	36A358006AAG01	2												
1, 2 SCR	36A358006AAG02		2	2	2									
1, 2 SCR	36A358006AAG03					2								
1, 2 SCR	36A358006AAG04						2	2						
1, 2 SCR	36A358006AAG05								2	2	2	2		
1, 2 SCR	36A358006AAG06												2	
M RLY	104X131AB001	1	1	1	1	1								
M RLY	104X131AB025						1	1	1	1	1	1	1	1
*M RLY	104X131AB035	1	1	1	1	1								
*M RLY	104X131AB026						1	1	1	1	1	1	1	1
IT	36A349214AAG01	1	1	1	1	1								
IT	36A349214AAG02						1	1	1	1	1	1	1	1
3, 4, 5 FU	104X109AD028	3	3	3	3	3	3	3	3	3	3	3	3	3
1, 2 FU	104X109AD035	1					2	2						
1, 2 FU	104X109AD036		1	1					2	2				
1, 2 FU	104X109AD037				1						2	2		
1, 2 FU	104X109AD074					1							2	
1, 2 FU	104X109AD056													2
*TA PC	6VSM01B1	1	1	1	1	1	1	1	1	1	1	1	1	1
DBR	6VSJ01					1								
DBR	6VSJ02		1	1	1									
DBR	6VSJ03	1									1	1	1	
DBR	6VSJ04						1	1	1	1				
DBR	6VSJ05													1
TOL	6VSL08						1							
TOL	6VSLI2							1						
TOL	6VSLI7	1							1					
TOL	6VSL25		1							1				
TOL	6VSL33			1										
TOL	6VSL37										1			
TOL	6VSL50				1							1		
TOL	6VSL75					1							1	
TOL	6VSLI00													1

*USED ON CONTROLLERS WITH FACTORY INSTALLED TIMED REFERENCE OPTION



SCR LEAD IDENTIFICATION: 1, 2SCR(C) IS THE BLACK LEAD WHICH ALSO HAS A RED WIRE SOLDERED AT THE SCR. 1, 2SCR(A) IS THE OTHER BLACK WIRE.

Figure 9. 6VHR Connection Diagram

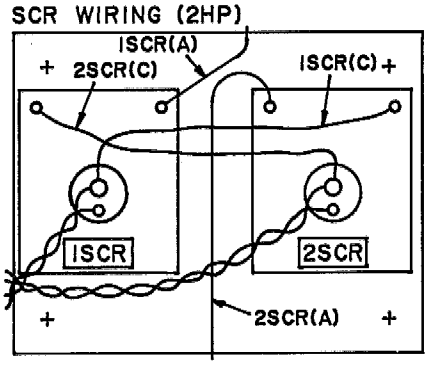


FIG. 9A

TOL MODIFICATION

DESCRIPTION

The TOL modification consists of a thermal overload relay, appropriate heater, and mounting screw. The thermal overload relay has a bi-metalic element which is exposed to heat generated by the motor armature current flowing in the TOL heater. The bi-metallic strip will cause the normally closed contacts in the TOL to open in the event of a prolonged motor overload.

CAUTION

THE TOL MODIFICATION DOES NOT PROVIDE MOTOR STALL PROTECTION. IF THE DRIVEN LOAD CAN STALL AN UNATTENDED MOTOR, OTHER MEANS OF THERMAL PROTECTION SHOULD BE USED.

INSTALLATION

Mount the TOL as shown in Figure 10 using the screw supplied. Disconnect the white wire at the "M" relay, point "B" and reconnect it to TOL point 2. Connect the loose white wire from TOL point 1 to "M" relay point "B" Disconnect the black wire from "M" relay point 6 and reconnect it to TOL point "Y". Connect the loose black wire from TOL point "X" to "M" relay point 6. To prevent breakage, be careful not to exert excessive force on the "M" relay terminals. This completes the TOL installation.

OPERATION

The TOL relay is provided with an adjustment knob for ambient temperature compensation. For normal 40°C ambient, this know should be set mid way.

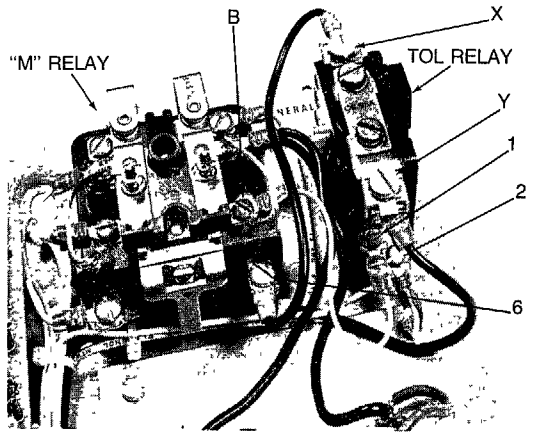


Figure 10. TOL Installation (1RC Removed for Clarity)

DBR MODIFICATION

DESCRIPTION

The dynamic braking option consists of a power resistor with two plug-in leads and mounting screws.

Normally closed relay contacts connect the resistor across the motor armature whenever the control station stop button is pressed. The motor then acts as a generator and dissipates its rotational energy as heat in the resistor.

INSTALLATION

Mount the DBR assembly as shown in Figure 11 using the two 6-32 thread cutting screws supplied. Plug the two wires from the DBR assembly into the Power Interface Board. This completes the DBR installation.

OPERATION

Press the "start" pushbutton and accelerate the motor to rated speed. Pressing the stop pushbutton should cause the motor to rapidly brake to zero speed.

CAUTION

HIGH START-STOP DUTY CYCLES OR EXTREMELY HIGH CONNECTED LOAD INERTIA MAY CAUSE THE MOTOR AND/OR DBR RESISTOR TO OVERHEAT. DO NOT EXCEED THE DBR RESISTOR 40 WATTS RMS RATING.

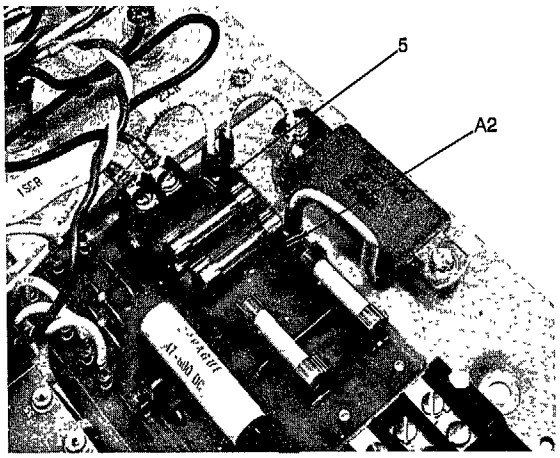
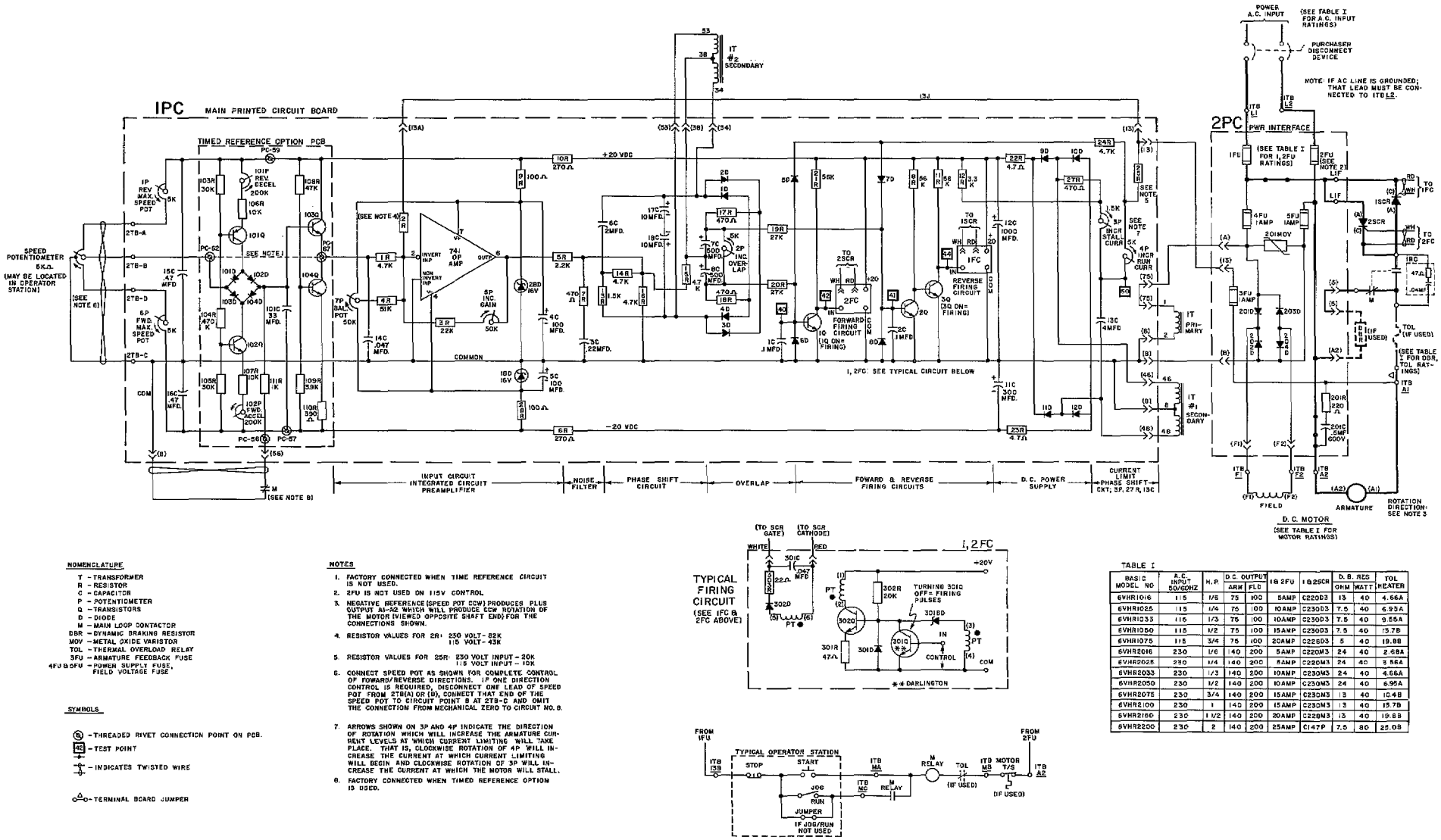


Figure 11. DBR Installation



NOMENCLATURE

- T - TRANSFORMER
- R - RESISTOR
- C - CAPACITOR
- P - POTENTIOMETER
- Q - TRANSISTORS
- D - DIODE
- M - MAIN LOOP CONTACTOR
- DBR - DYNAMIC BRAKING RESISTOR
- MOV - METAL OXIDE VARISTOR
- TOL - THERMAL OVERLOAD RELAY
- 3FU - ARMATURE FEEDBACK FUSE
- 4FU & 5FU - POWER SUPPLY FUSE, FIELD VOLTAGE FUSE

SYMBOLS

- (S) - THREADED RIVET CONNECTION POINT ON PCB.
- 46 - TEST POINT
- INDICATES TWISTED WIRE
- TERMINAL BOARD JUMPER

NOTES

1. FACTORY CONNECTED WHEN TIME REFERENCE CIRCUIT IS NOT USED.
2. 2FU IS NOT USED ON 115V CONTROL
3. NEGATIVE REFERENCE (SPEED POT CCW) PRODUCES PLUS OUTPUT A1-A2 WHICH WILL PRODUCE CCW ROTATION OF THE MOTOR (VIEWED OPPOSITE SHAFT END) FOR THE CONNECTIONS SHOWN.
4. RESISTOR VALUES FOR 2R: 230 VOLT - 82K; 115 VOLT - 43K
5. RESISTOR VALUES FOR 25R: 230 VOLT INPUT - 20K; 115 VOLT INPUT - 10K
6. CONNECT SPEED POT AS SHOWN FOR COMPLETE CONTROL OF FOWARD/REVERSE DIRECTIONS. IF ONE DIRECTION CONTROL IS REQUIRED, DISCONNECT ONE LEAD OF SPEED POT FROM 2TB(A) OR (D), CONNECT THAT END OF THE SPEED POT TO CIRCUIT POINT 8 AT 2TB-C AND OMIT THE CONNECTION FROM MECHANICAL ZERO TO CIRCUIT NO. 8.
7. ARROWS SHOWN ON 3P AND 4P INDICATE THE DIRECTION OF ROTATION WHICH WILL INCREASE THE ARMATURE CURRENT LEVELS AT WHICH CURRENT LIMITING WILL TAKE PLACE. THAT IS, CLOCKWISE ROTATION OF 4P WILL INCREASE THE CURRENT AT WHICH CURRENT LIMITING WILL BEGIN AND CLOCKWISE ROTATION OF 3P WILL INCREASE THE CURRENT AT WHICH THE MOTOR WILL STALL.
8. FACTORY CONNECTED WHEN TIMED REFERENCE OPTION IS USED.

TABLE I

BASIC MODEL NO	A.C. INPUT 50/60HZ	H.P.	D.C. OUTPUT ARM	D.C. OUTPUT FLD	1B 2FU	1B2SCR	D.B. RES OHM	D.B. RES WATT	TOL HEATER
6VHR1016	115	1/6	75	100	5AMP	C220D3	13	40	4.66A
6VHR1025	115	1/4	75	100	10AMP	C230D3	7.5	40	6.95A
6VHR1033	115	1/3	75	100	10AMP	C230D3	7.5	40	9.55A
6VHR1050	115	1/2	75	100	15AMP	C230D3	7.5	40	13.7B
6VHR1075	115	3/4	75	100	20AMP	C226D3	5	40	19.8B
6VHR2016	230	1/6	140	200	5AMP	C220M3	24	40	2.68A
6VHR2025	230	1/4	140	200	5AMP	C220M3	24	40	3.56A
6VHR2033	230	1/3	140	200	10AMP	C230M3	24	40	4.66A
6VHR2050	230	1/2	140	200	10AMP	C230M3	24	40	6.95A
6VHR2075	230	3/4	140	200	15AMP	C230M3	13	40	10.4B
6VHR2100	230	1	140	200	15AMP	C230M3	13	40	13.7B
6VHR2150	230	1 1/2	140	200	20AMP	C226M3	13	40	19.8B
6VHR2200	230	2	140	200	25AMP	C147P	7.5	80	25.0B

Figure 12. 6VHR Statotrol Drive Schematic Diagram

WARRANTY

The Contract Warranty for Statotrol drives is stated in the General Electric Apparatus Handbook, Section 98, page 11. The purpose of the following is to provide specific instructions to the Statotrol user regarding warranty administration and how to obtain assistance on out-of-warranty failures.

STATOTROL CONTROLLERS (1/6 – 2 HP)

The warranty covers all major parts of the controller such as the main printed circuit board, modification kits, contactors, etc., but does not provide for replacement of fuses or the complete controller.

1. In the event of failure or misapplication during "in warranty", refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

   STATOTROL DRIVE SERVICE
   Erie, Pennsylvania
   (814) 455-0521

   Before calling, list catalog numbers of the controller, motor, operator's station and any optional modifications. Include orange colored four digit identification number located in lower left corner of controller chassis.

3. Statotrol Service will ship prepaid, at no charge, any parts that have failed in warranty and advise if failed part(s) should be returned to the factory. The factory address is:

   General Electric Company
   Speed Variator Products Section, DCM&GPD
   1100 Lawrence Parkway
   Erie, Pennsylvania 16531
   Attention: STATOTROL DRIVES SERVICE

4. For parts, contact your local General Electric Sales Office or your "Authorized" General Electric Parts Distributor.

"Exchange" parts at "Exchange" prices may be available for selected items. When returning parts for exchange, pack them carefully to prevent damage in transit.

STATOTROL MOTORS

Statotrol motor repairs are generally handled by General Electric Small Motor Service Stations or Apparatus Service Shops. For specific instructions on your motor, call STATOTROL DRIVE SERVICE at the number listed above and furnish complete nameplate data.

NOTES

NOTES

GENERAL ELECTRIC COMPANY – DC MOTOR & GENERATOR DEPARTMENT
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK–47612A (8–78) 2M(F)